7 ET 8 OCTOBRE 2026

INSTITUT DE RECHERCHE D'HYDRO-QUÉBEC (IREQ)

VARENNES (QUÉBEC) | EN PRÉSENTIEL ET SANS FRAIS

SYMPOSIUM 3i pour le système énergétique de demain

Supporting Grid Recovery Through Resilient Wireless Connectivity

Selective updates of a wireless digital twin to support field monitoring and disaster recovery

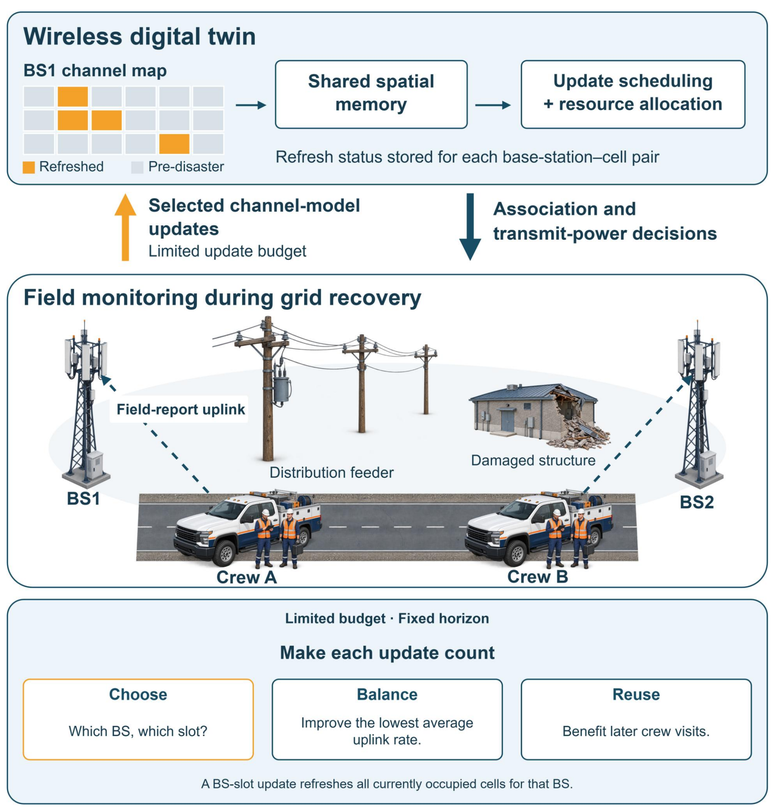
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Operators need field information → **Field teams need reliable connectivity** → **Disaster damage changes propagation** → **The channel map becomes stale** → **Selective updates guide BS association and transmit power**

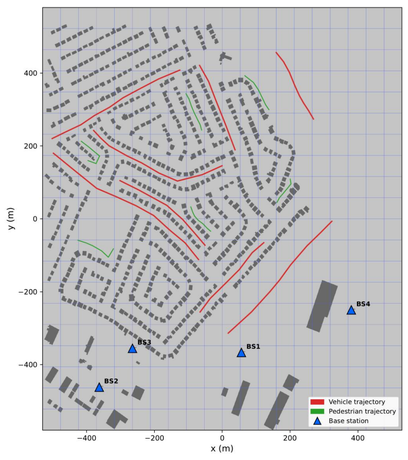
1 PHYSICAL PROBLEM

Field teams send reports for grid recovery [1]. Damage can leave the wireless digital twin's channel map stale and misguide wireless decisions [2, 3].



A wireless digital twin for recovery uplinks

Selective updates refresh the wireless digital twin's channel map after propagation changes. The refreshed knowledge guides association and transmit power for field-report uplinks.



Ray tracing



Ray tracing translates scene geometry into the wireless digital twin's channel map.

As users move, a cell refreshed for one base station can benefit later visits—including visits by another user. Shared spatial memory preserves this knowledge during the fixed post-disaster horizon.

Sionna RT evaluation [4]

Parameter	Setting
Location / scene	1 × 1 km suburban district near Montréal
Carrier	3.5 GHz
Materials	Concrete ground; brick walls; metal roofs
Network	4 base stations; 12 users
Time horizon	40 slots; 0.5 s per slot
Spatial grid	20 × 20 cells

System model: uplink rate

$$\widehat{R}_i[t] = \log_2\left(1 + \sum_b a_{ib}[t]\widehat{\gamma}_{ib}[t]\right),$$

$$\widehat{\gamma}_{ib}[t] = \frac{p_i[t]\widehat{g}_{ib}[t]}{\sigma^2 + I_b[t]},$$

$$I_b[t] = \sum_{b' \neq b} \sum_j a_{jb'}[t] p_j[t] \widehat{g}_{jb}[t].$$

OFDMA removes intra-cell interference. The denominator includes inter-cell interference and noise. Actual rate uses Scene-B channels.

2 FORMULATION & SOLUTION

Joint scheduling and resource allocation

Which BS should update—and when—under a limited update budget?

Goal: maximize the worst user's time-averaged rate.

$$\max_{\mathbf{a},\mathbf{p},\mathbf{s},\mathbf{m}} \ \eta \qquad \text{s.t.} \quad \frac{1}{T}\sum_{t=1}^{T} \widehat{R}_i[t] \geq \eta, \quad \forall i$$

a: association p: transmit power s: DT update
m: spatial memory R̂: predicted uplink rate

The digital twin selects updates, user association, and transmit power to maximize the worst user's time-averaged uplink rate.

Update budgets and resource limits

$$\sum_b s_b[t] \leq C, \qquad \sum_{t,b} s_b[t] \leq S_{\text{tot}},$$

$$\sum_b a_{ib}[t] \leq 1, \qquad \sum_i a_{ib}[t] \leq N_b,$$

$$0 \leq p_i[t] \leq P_{\max}, \qquad a_{ib}[t], s_b[t] \in \{0,1\}.$$

Persistent spatial memory

$$m_{b,c}[t] = \max\left\{m_{b,c}[t-1],\ s_b[t]\mathbf{1}_{\{c \in \mathcal{V}[t]\}}\right\},$$

$$\widehat{g}_{ib}[t] = g^A_{ib}[t] + m_{b,c_i[t]}[t]\left(g^B_{ib}[t] - g^A_{ib}[t]\right).$$

A BS-slot action refreshes all currently occupied cells for that BS. V[t] is the occupied-cell set.

1 Linear memory reformulation

$$\mathcal{T}_{it} = \{\tau \leq t : c_i[t] \in \mathcal{V}[\tau]\},$$

$$q_{ib}[t] \geq s_b[\tau] \quad (\forall \tau \in \mathcal{T}_{it}), \qquad q_{ib}[t] \leq \sum_{\tau \in \mathcal{T}_{it}} s_b[\tau],$$

$$\widehat{g}_{ib}[t] = g^A_{ib}[t] + q_{ib}[t]\left(g^B_{ib}[t] - g^A_{ib}[t]\right).$$

The history includes visits by any user to the current cell; q represents updated-cell status.

2 Successive convex approximation

$$\widehat{R} = \log_2(D+S) - \log_2 D,$$

$$\widehat{R}^{(r)}_{\text{lb}} = \log_2(D+S) - \log_2 D^{(r)} - \frac{D - D^{(r)}}{D^{(r)} \ln 2}.$$

S: useful signal; D: interference + noise. Linearizing $\log_2 D$ gives a concave rate lower bound and a convex subproblem for each block.

Two solution methods

Trajectory-based

Schedule once

Maximize number of updated cells

Fixed DT update schedule

Association ↔ Transmit power

Repeat until convergence

Channel-aware

Post-disaster channel information

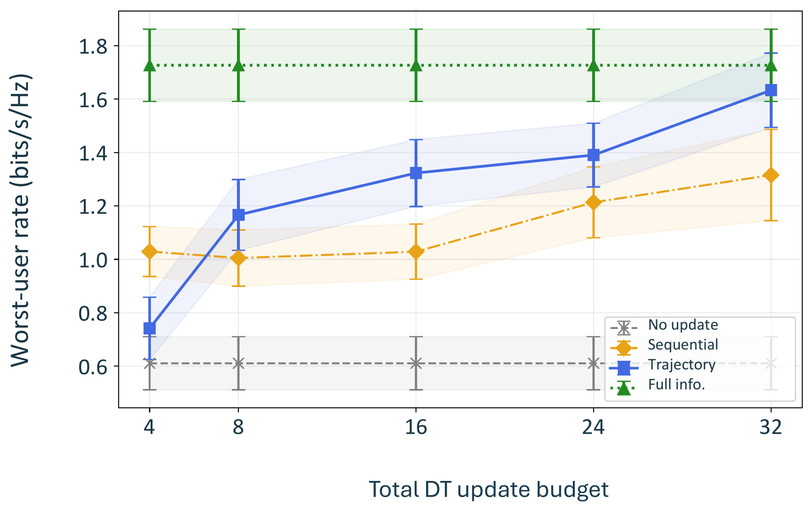
DT update
Recompute digital-twin channels (ray tracing)

Association → Transmit power

Repeat until convergence

3 EVIDENCE & DESIGN

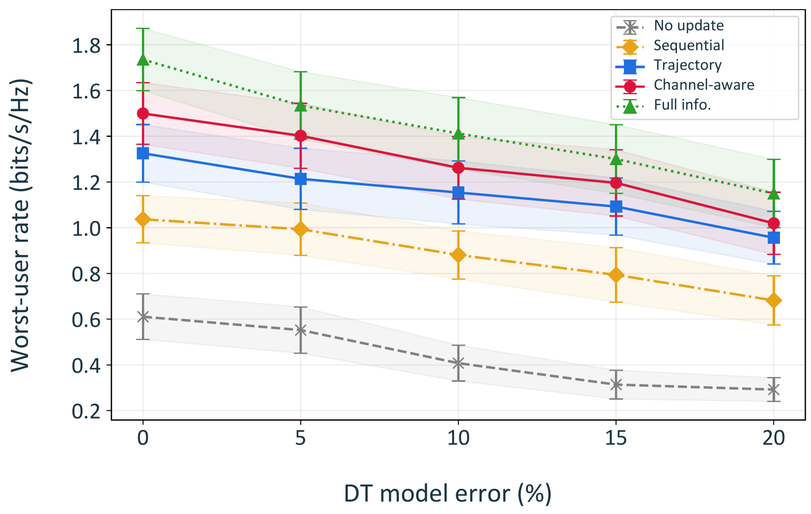
More DT updates close the performance gap



45% → 93% gap closed

Trajectory-based method, 8 → 32 updates. At 4 updates, sequential coverage performs better.

Benefits persist with model error

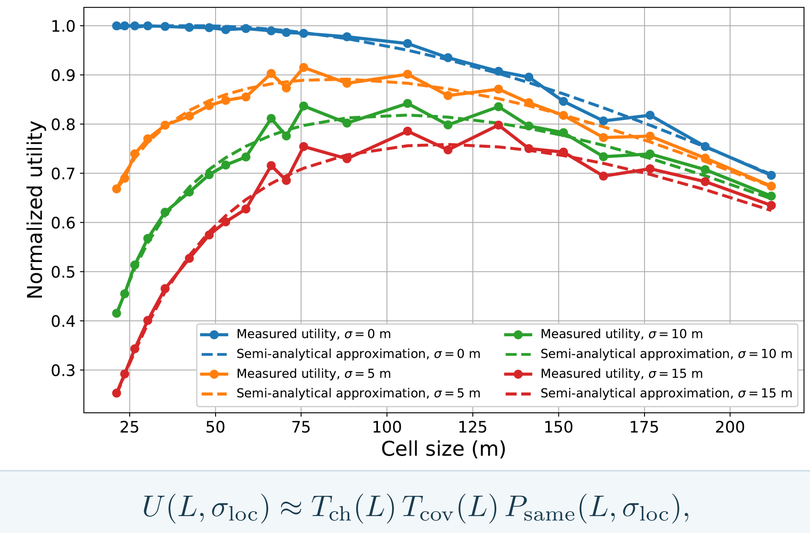


72% of nominal rate retained

Trajectory-based method at 20% error. "Full info." also uses perturbed estimates in this sweep.

Spatial cell size and localization uncertainty

Smaller spatial cells are not always better under localization uncertainty.



$$U(L,\sigma_{\text{loc}}) \approx T_{\text{ch}}(L)\, T_{\text{cov}}(L)\, P_{\text{same}}(L,\sigma_{\text{loc}}),$$

$$T_{\text{ch}}(L) = e^{-\lambda_{\text{ch}}(c_0 + c_1 L + c_2 L^2)}, \qquad T_{\text{cov}}(L) = \left(1 - e^{-\gamma_1 L^{\gamma_2}}\right)^{\gamma},$$

$$P_{\text{same}}(L,\sigma_{\text{loc}}) = I(L,\sigma_{\text{loc}})^2,$$

$$I(L,\sigma_{\text{loc}}) = \text{erf}\left(\frac{L}{\sqrt{2}\sigma_{\text{loc}}}\right) + \frac{\sigma_{\text{loc}}}{L}\sqrt{\frac{2}{\pi}}\left[e^{-L^2/(2\sigma^2_{\text{loc}})} - 1\right].$$

Selected references

[1] M. Qanbaryan, S. Y. Derakhshandeh, and Z. Mobini, "UAV-enhanced damage assessment of distribution systems in disasters with lack of communication coverage," SEGAN, vol. 33, 100984, 2023.
[2] H. X. Nguyen et al., "Digital Twin for 5G and Beyond," IEEE Commun. Mag., vol. 59, no. 2, pp. 10–15, 2021.
[3] A. Tanaka, B. Du, and R. Li, "Digital Twin-Empowered Post-Disaster Movable Base Station Deployment Optimization," IEEE Commun. Mag., vol. 63, no. 12, pp. 38–44, 2025.
[4] J. Hoydis et al., "Sionna RT: Differentiable Ray Tracing for Radio Propagation Modeling," IEEE GLOBECOM Workshops, pp. 317–321, 2023.
Our paper: Joint Digital Twin Synchronization Scheduling and Resource Allocation for Post-Disaster Wireless Networks • IEEE TNSM, 2026 • doi: 10.1109/TNSM.2026.3714917

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